

Binomial distribution

$$P(K = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$\binom{n}{k} = \frac{n!}{k! (n - k)!}$$

Poisson process

$$P\{N(t) = k\} = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, t \geq 0$$

Queuing theory

Average number of customers arriving in one unit of time	λ
The average number of customers who can be served per time period	μ
Server utilisation	$\rho = \frac{\lambda}{\mu}$
Average number of customers in the system.	$L = \frac{\rho}{1 - \rho}$
Average number of customers in the queue.	$L_Q = \frac{\rho^2}{1 - \rho}$
Average time a customer spends in the system.	$W = \frac{1}{\mu - \lambda}$
Average time a customer spends in the queue.	$W_Q = \frac{\lambda}{\mu(\mu - \lambda)}$
Probability that there are no customers in the system	$p_0 = 1 - \rho$
Costs for one customer in system per time unit	CC
Costs for one server per time unit	CS
Average number of customers in the system.	L
Number of servers	m
Total average costs	TC
$TC = CC * L + CS * m$	